Graphical Models, Conditional Independence, and Message Passing · Quiz explanations

Original checkpoint explanations aligned with the W2 lecture. These are not project solutions.

1. What must be specified before the expression “perform inference” defines a mathematical task?

Specify the model/parameters, observed variables and values, query variables, nuisance variables and desired output (for example a normalized marginal, joint MAP, evidence probability or expectation). Conditioning and summation must be explicit.

2. Why does observing the middle node block a chain but potentially activate a collider?

In a chain, the factorization conditioned on the middle separates the endpoint factors. In a collider, conditioning introduces a likelihood involving both parents, which generally couples them. Observing a collider descendant can also activate the path.

3. Does failure of d-separation prove dependence for every parameter choice?

No. It means the graph does not guarantee independence. Special parameters can create additional independences. D-separation implies independence for distributions that factorize over the DAG; the converse for a particular distribution needs stronger assumptions.

4. Why is a message from i to j indexed by the state of j?

It sums out the sender-side variables while holding the boundary variable at the recipient fixed. The result summarizes how that entire component weights each possible recipient state.

5. In the worked chain, why is the product [0.2356, 0.2508] not yet a posterior distribution?

Its entries sum to the evidence probability 0.4864, not one. Dividing by that sum gives [0.484375, 0.515625]. Arbitrary message scaling preserves normalized marginals but requires scale bookkeeping to recover evidence mass.

6. Can independently chosen marginal modes differ from joint MAP?

Yes. In the displayed joint table the joint maximum is (0,0), while the marginal modes are A=1 and B=0. Summation and maximization answer different queries; marginal decisions can also violate joint constraints.

7. What is the difference between filtering and smoothing, and why does it matter for evaluation?

Filtering uses observations only through the current time. Smoothing additionally uses future observations. Reporting smoothing performance as online filtering performance leaks information unavailable to the online system.

8. What can a small loopy-BP message residual establish, and what can it not establish?

It indicates that consecutive normalized messages have changed little under the chosen update rule and tolerance. It does not certify exact posterior marginals, a globally correct joint distribution, convergence from all initializations or model validity.